Active Recovery Is Not Just for Athletes: Employees Need to Train for Recovery to Enhance Work

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The data analyzed here were collected and deposited by Peuckmann et al. (2026; Open Science Framework, https://osf.io/32rhy/), and the depositing team has published two papers from them (Peuckmann et al., 2026; Reis et al., 2024). This article reports a secondary analysis of that public, de-identified deposit, and no new data were collected. Supplemental material for this article, including the plan amendments, the run record, and the full automation record, is available at https://osf.io/r68v7/. These analyses have not been presented, posted, or published elsewhere.

The author has no connection to the depositing team and declares no conflict of interest. This research received no specific funding. During the preparation of this work the author used large language models accessed through Cursor and the Claude application (Claude, Grok, GPT, Kimi, Gemini, and Composer; each is attributed to its vendor in the Method) to draft and revise text, write and review analysis and figure code, and check reported statistics; a word processor with Clippy and autocorrect; and statistical software, so that no coefficient was obtained by hand. The author reviewed, edited, and verified the output and takes full responsibility for the content. These tools are not authors.

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Abstract

Trials of recovery training report mean gains in psychological detachment, but the mean does not show whether training helped the employees who detached least or those who already detached well. An aptitude-treatment account predicts larger gains at the low end of the baseline distribution (compensation), and a cumulative-advantage account predicts larger gains at the high end (accumulation). I tested the two accounts in a prespecified reanalysis of a randomized trial among German employees ($N = 393$, 76% women, mean age 41; 247 with posttest data) that compared a wait-list control with a 6-week cognitive-behavioral recovery training and a 6-week mindfulness-based training. The primary test was the baseline × condition interaction in a Lin-adjusted regression of posttest detachment; secondary analyses were quantile treatment effects, changes-in-changes and Lee bounds, multiple imputation, and a missingness model. Both trainings raised detachment at the mean baseline by about 0.9 points, and quantile effects were larger at the .25 than at the .75 quantile in both groups. The cognitive-behavioral interaction was −0.21 (95% CI [−0.50, 0.08]), leaning toward compensation, and the mindfulness interaction 0.04 (95% CI [−0.24, 0.33]); the condition contrast was 0.25 (95% CI [−0.01, 0.51]), and all three intervals included zero. Dropout before the posttest was not detectably patterned by baseline detachment in any group. The results support neither compensation nor accumulation. Neither training detectably favored employees at either end of the baseline distribution, and the intervals bound how large either pattern could be in a trial of this size.

Keywords: recovery training, psychological detachment, quantile treatment effects, aptitude-treatment interaction, secondary data analysis

Active Recovery Is Not Just for Athletes: Employees Need to Train for Recovery to Enhance Work

Employees who do not detach from work during off-job time report poorer well-being (Sonnentag & Fritz, 2007). Trainings that teach people how to detach have become a common object of intervention research in occupational health psychology (Hahn et al., 2011; Karabinski et al., 2021). Athletes treat recovery as part of training, something scheduled and practiced; employees are mostly left to recover as best they can, and recovery trainings are an attempt to change that. A meta-analysis of 30 studies found an average effect on detachment of $d = 0.36$, with larger effects for longer and higher-dose programs (Karabinski et al., 2021). The evidence base is almost entirely a record of mean differences. A mean difference shows whether the average employee detached more after training than the average control participant. A mean difference leaves open whether the gain accrued mainly to employees who detached least to begin with or to employees who were already detaching reasonably well. That is the question for anyone deciding whom to offer a training.

Peuckmann et al. (2026) invoke the recovery paradox, the observation that people who most need recovery tend to engage least in the activities that produce it (Sonnentag, 2018). A training that reaches the people the paradox describes should narrow the distribution of detachment. A training whose exercises presuppose a capacity those people lack may instead widen the distribution while still raising the mean. Both outcomes are consistent with a positive mean effect, so separating them requires the distribution of detachment, not the mean alone.

Two Accounts of Whom Training Helps Most

Two established lines of theory make opposite predictions about this pattern. The first is the aptitude-treatment interaction tradition (Cronbach & Snow, 1977; Snow, 1991), which in

industrial-organizational psychology appears as the study of how trainee characteristics moderate training outcomes (Gully et al., 2002). On a compensatory reading, a training that supplies what the worst-off lack benefits them most. An employee who has never planned off-job time or set a boundary around it has more to gain from being taught how than an employee who already does both. Applied to a recovery training, the account yields the prediction tested here, namely a negative association between baseline standing and gain, and a posttest distribution tighter than the control distribution.

The second line is cumulative advantage (DiPrete & Eirich, 2006; Merton, 1968). It has been applied to workplaces as a general model of how initial differences in resources produce further differences (van Dijk et al., 2020). It has also been applied to interventions, where universal programs have been found to widen gaps because advantaged groups gain more even when disadvantaged groups also gain (Ceci & Papierno, 2005; Lorenc et al., 2013). On this reading, a training whose use requires a capacity the better-off already possess benefits them most. Mindfulness practice is a plausible case. Sitting with experience without reacting to it is itself a regulatory act, and people who already regulate well may take more from instruction in it. Here the expected association between baseline standing and gain is positive, with a posttest distribution wider than the control distribution.

These accounts differ in the mechanism they posit, and the trial analyzed here supplies a design contrast that the two accounts treat differently. The cognitive-behavioral training taught a skill, the planning and boundary management of off-job time, and drew on the DRAMMA model of leisure, which names detachment-recovery, autonomy, mastery, meaning, and affiliation as the psychological needs that leisure serves (Newman et al., 2014, as cited in Peuckmann et al., 2026). The mindfulness training taught a stance, acceptance and decentering, and drew on

mindfulness-based stress reduction and mindfulness-based cognitive therapy (Peuckmann et al., 2026). The compensatory account predicts compression most confidently for the skill condition. The accumulation account predicts spread most confidently for the stance condition, and, if effortful planning itself draws on self-regulatory capacity, spread for the skill condition as well. The present study is a test between the two accounts, condition by condition, rather than a confirmation of either.

What Prior Work Has and Has Not Shown

Recovery research has asked this question with other methods, and it remains open. Detachment interventions had larger mean effects among participants with initial health or recovery impairments (Karabinski et al., 2021), a meta-analytic moderation that points toward compensation but is not a within-trial test of heterogeneity. A test of baseline moderation in a low-dose mindfulness intervention found none (Hülsheger et al., 2015). A mindfulness segmentation training raised detachment more among employees with low segmentation preference (Althammer et al., 2021), a moderator conceptually close to baseline detachment but not the same variable. A study of police officers asked for whom mindfulness training works and tested personality and social-norm moderators (Krick & Felfe, 2020). More broadly, it has been argued that intervention effects in behavioral science are heterogeneous as a rule (Bolger et al., 2019; Bryan et al., 2021) and that heterogeneity should be studied through preregistered, theory-driven moderation rather than discovered post hoc (Bryan et al., 2021). The same argument has been made for organizational interventions under the heading of what works for whom (Nielsen & Miraglia, 2017). Program evaluation has made a parallel point: a mean impact can be identical across programs whose effects on the distribution differ entirely (Bitler et al., 2006).

Among the studies reviewed above, none has applied distributional estimators to a recovery-training trial, and none has set compensation and accumulation against each other as prespecified competing hypotheses in the same design. The deposited trial permits a distributional analysis and a prespecified contrast of compensation with accumulation. It randomized 393 employees to three conditions, and its two trainings differ on the dimension the accounts need.

The Trial and Its Two Published Analyses

The deposited data come from a randomized trial (Peuckmann et al., 2026) in which employees who volunteered for recovery training were assigned to a wait-list control (n = 132), a 6-week online cognitive-behavioral recovery training (CBT; n = 131), or a 6-week online mindfulness-based training (MBT; n = 130). Psychological detachment, relaxation, control, and mastery were measured before and after the training phase. The trial has been analyzed twice by its investigators. The first analysis compared the two trainings on mean detachment with latent change models and Bayes-factor equivalence tests and concluded that they raised detachment to a similar extent (Reis et al., 2024). The second pooled the two trainings and asked whether profiles of the four recovery experiences, and transitions between them, changed with training (Peuckmann et al., 2026).

Neither analysis addresses the question asked here. The first is a statement about means, and the second is a statement about profile membership in a pooled training group. Whether each training, kept separate, acted more strongly at the bottom or the top of the detachment distribution, and whether the employees lost to the posttest were the ones who started lowest, are questions the deposited data can answer and the published analyses do not.

The Present Study

The present study is a prespecified reanalysis of the deposited trial, guided by four research questions. The first two ask whether, relative to control, CBT (RQ1) and MBT (RQ2) acted more strongly in the lower or the upper part of the posttest detachment distribution. The third asks whether the two trainings differed in where on the distribution they acted (RQ3). The fourth asks whether dropout before the posttest was patterned by baseline detachment and whether that pattern differed by condition (RQ4).

Each hypothesis is stated as a competing pair. Both accounts predict that the size of the gain depends on where a person starts. The primary test for H1 and H2 is therefore the baseline × condition interaction in a regression of posttest detachment on condition, centered baseline detachment, and their products (Lin, 2013). The marginal quantile treatment effect (Firpo et al., 2009) is the secondary quantity in the same plan; it locates where on the posttest distribution any change registered.

Hypothesis 1 (CBT, the skill condition). Compensation: the baseline × CBT interaction is negative, so that the training's effect is larger for employees with lower baseline detachment. Accumulation: the interaction is positive.

Hypothesis 2 (MBT, the stance condition). Compensation: the baseline × MBT interaction is negative. Accumulation: the interaction is positive. The accumulation account makes its prediction more confidently for MBT than for CBT.

Hypothesis 3 (condition contrast). If each account is right about its own condition, the MBT interaction exceeds the CBT interaction. If accumulation holds for both conditions, the two interactions share a sign and the contrast is indeterminate, the plan's label for an interval that covers zero.

Hypothesis 4 (reach). Compensation implies that CBT retains low-baseline employees at least as well as the control condition does. Accumulation implies that low-baseline employees are lost from the training groups at a higher rate than from control. The direction was left open, and the sign of the baseline × condition association in a missingness model is the finding.

Method

Transparency and Analysis Plan

This is a secondary analysis of an existing, publicly deposited dataset, with an analysis plan fixed before any planned statistic was computed on the outcome data. It is reported in the manner recommended for such analyses (van den Akker et al., 2021; Weston et al., 2019). The study plan fixed the hypotheses, the quantities to be estimated, the decision rules, the bootstrap settings, the seed, the multiplicity adjustment, and the claim boundaries. It was frozen on 22 October 2026, before any posttest statistic was computed, and every planned estimator was then fitted once, in a single prespecified run with no refitting and no data-dependent choice. The supplemental material gives the dated plan amendments, none of which changed a quantity to be estimated, a decision rule, the seed, or the bootstrap specification, together with the run record, the prior exposure to column-level aggregates of the deposited file, and the full record of software and automation.

Participants and Procedure

Design facts are taken from the primary report (Peuckmann et al., 2026), which gives the full protocol. The trial was conducted in Germany by a team at Saarland University, was registered as a clinical trial (DRKS00024933), and was approved by an ethics committee that the primary report does not name. Participants gave informed consent at the start of the intake questionnaire. The present analysis used only the public, de-identified deposit, did not contact

participants, and was not a new study of human participants. Sequence generation and allocation are described in the primary trial reports (Peuckmann et al., 2026; Reis et al., 2024). Participants were necessarily aware of their condition, and all outcomes were self-reported. Participants were recruited through media announcements and workplace health-management offices as volunteers for a training promoting recovery from work. Inclusion criteria were age over 18, core working hours between 6:45 a.m. and 5:30 p.m., a smartphone with internet access, and no current psychotherapy; the last is an exclusion rule, not a record of diagnoses among those enrolled. The 393 participants who completed the intake questionnaire had a mean age of 40.56 years (*SD* = 10.94), and 75.57% were women. The largest occupational groups were health care and education (25.45%) and administration (14.76%), and contracts averaged 35.66 hours per week. The primary report gives age, gender, occupation, and contract hours. Sex as distinct from gender, racial or ethnic identity, nativity, socioeconomic standing beyond occupation, disability status, and sexual orientation were not reported there and are not in the deposit, so they cannot be described here. Demographics by condition and reasons for dropout are likewise not part of the deposit as documented.

The protocol spanned 8 weeks: a questionnaire before the training phase (the pretest, following the usage of Reis et al., 2024), a week of experience sampling, 6 weeks of training for the two training groups (one self-guided online module released per week), a further week of experience sampling, and a questionnaire after the training phase (the posttest). The pretest detachment score is called baseline where it enters a model as a covariate. MBT drew on mindfulness-based stress reduction and mindfulness-based cognitive therapy; CBT drew on the DRAMMA model and taught planning and boundary-management techniques for off-job time.

In the pooled training group, 63% had completed the week-six module (Peuckmann et al., 2026). The wait-list control received the training after the study.

Group sizes in the deposited file were 132 (control), 131 (CBT), and 130 (MBT). Posttest recovery scores were present for 247 participants (62.8%): 92 in control (69.7%), 79 in CBT (60.3%), and 76 in MBT (58.5%). These 247 participants are the posttest sample. The two follow-up detachment parcels were present for 204 participants (68, 67, and 69). No participant was excluded for any reason other than a missing outcome; the analysis sample for each estimator is stated with the estimate.

Measures

Psychological detachment, relaxation, control, and mastery were measured before and after the training phase with the four-item subscales of the Recovery Experience Questionnaire (Sonnentag & Fritz, 2007). Items carried the time frame “in the last two weeks” and were rated from 1 (*does not apply at all*) to 7 (*fully applies*). The deposited scores are item means, so each lies on a quarter-point grid from 1 to 7 (25 possible values). McDonald’s ω was .87 (pre) and .92 (post) for detachment, with a pre–post correlation of .48, and between .78 and .90 for the other three scales (Peuckmann et al., 2026, Table 1). Example items are “I didn’t think about work at all” (detachment) and “I did relaxing things” (relaxation). Posttest detachment is the primary outcome, baseline detachment is the covariate, and relaxation, control, and mastery are secondary outcomes.

Sleep quality and perceived stress were deposited as standardized residualized-change scores (mean 0, standard deviation 1) rather than as raw scale scores. Because their spread is fixed by construction, they are analyzed as location contrasts only and are not entered into any distributional estimator.

Condition is the deposited three-level assignment variable. *Follow-up detachment* is the mean of the two follow-up parcels for the 204 participants with both, used in one descriptive repeat and nowhere else. The follow-up scale means themselves were not deposited.

Analytic Strategy

Quantities Estimated and Decision Rule

Family A, the primary family, comprised the baseline × CBT and baseline × MBT interactions in the Lin (2013) model and the H4 baseline × condition interactions in the missingness model. The Lin model is ordinary least squares of posttest detachment on two condition indicators, baseline detachment centered at the posttest-sample mean, and the two products, with heteroskedasticity-consistent standard errors of the HC2 type. For H1 and H2, compensation is supported if the 95% confidence interval for the interaction lies entirely below zero, accumulation if entirely above, and the result is labeled indeterminate otherwise; the sign of the point estimate was reported but does not decide. H3 is the linear contrast MBT minus CBT on the same fit, read as supportive of the account-specific prediction if its interval lies entirely above zero, contrary if entirely below, and indeterminate otherwise. The 99% interval was reported beside every 95% interval; only the 95% interval decides a label. Within Family A, three *p* values entered a Holm adjustment: the two HC2 Wald *p* values for the interactions and one 2-*df* Wald *p* for the pair of baseline × condition terms in the missingness model. Family B, the secondary family, comprised all remaining analyses below and was reported unadjusted and labeled secondary. Every planned estimate was reported whether or not its interval excluded zero.

Interpretive Boundaries

Two interpretive boundaries were fixed in the plan. First, baseline detachment is the baseline value of the outcome, not a measure of self-regulatory capacity, and no capacity measure exists in the deposited file. A baseline-conditional effect is therefore read as such, and nothing is inferred about capacity from baseline rank. Second, quantile treatment effects identify changes in marginal distributions, not individual gains (Ding et al., 2016). Quantile results are therefore described as a rise at the lower quantile, not as larger gains for employees who started low, unless the baseline-conditional interaction supports the latter.

Mean Benchmark

The condition coefficients of the Lin model are the effect of each training at the mean baseline, and they are the benchmark against which the prespecified contribution criterion was read. The contribution criterion is whether any distributional result changes the conclusion the mean contrast supports, including the conclusion published for this trial (Reis et al., 2024). That reading is reported with the distributional results, under Summary of Distributional and Bounds Evidence. Lin adjustment with treatment-covariate interactions estimates the average treatment effect consistently in a randomized experiment whether or not the linear model is correct, and does not reduce asymptotic precision relative to the unadjusted difference in means (Lin, 2013). The single baseline measurement makes analysis of covariance the appropriate form (McKenzie, 2012).

Quantile Treatment Effects

For each training group and each τ in {.25, .50, .75}, the unconditional quantile treatment effect QTE(τ) is the difference between the type-7 sample quantiles (linear interpolation between adjacent order statistics) of posttest detachment in the training group and in the control group, in

the posttest sample (Firpo et al., 2009). The gradient $G = \mathrm{QTE}(.75) - \mathrm{QTE}(.25)$ summarizes where the effect is larger; a negative gradient means larger gains at the lower quantile. Quantiles .10 and .90 were reported descriptively because about eight participants lie in each tail cell. The recentered-influence-function regression with condition as the only regressor, and the version adjusted for centered baseline and its products, were reported beside the raw quantile difference. Inference was by nonparametric bootstrap, resampling participants within condition, $B = 2{,}000$, seed 20261022, with percentile intervals as the decision interval and bias-corrected and accelerated (BCa) intervals as a sensitivity. Because the outcome is discrete, the number of distinct bootstrap values of each statistic was reported as a diagnostic, and a kernel-smoothed quantile (Gaussian kernel, Silverman bandwidth; halved and doubled as sensitivities) was reported beside the type-7 value. The number of distinct support values between the .25 and .75 quantiles in each group was reported so that the reader can see how coarse the grid is where the gradient is read.

Changes-in-Changes

The changes-in-changes estimator (Athey & Imbens, 2006) forms the counterfactual posttest distribution of a training group from the control group's own pre-to-post movement at each rank. Because the outcome lies on a 25-point grid, the continuity assumption fails. The discrete-outcome lower and upper bounds (Athey & Imbens, 2006, Theorem 4.1) were therefore reported as the estimate, with a point estimate from uniform jittering within grid cells (±0.125, 50 draws) as a sensitivity. The primary analysis population was the panel of 247 participants with both measurements, with the repeated-cross-section form (all 393 at baseline) as a sensitivity. The control group's pre-to-post quantile movement was reported so that the reader can see how much of any training-group result is the control counterfactual itself. A sign

disagreement between changes-in-changes and the quantile treatment effects was reported, and the two estimates were not reconciled.

Attrition and Bounds

H4 was tested with a logistic regression of posttest missingness on centered baseline detachment, condition indicators, and their products (HC2 standard errors), using all 393 participants. Coefficients were reported per scale point and per baseline standard deviation, and the baseline distributions of participants who did and did not provide a posttest score were compared within each group by a two-sample Kolmogorov–Smirnov test, reported descriptively. Because retention was higher in the control group than in either training group, the contrast within the posttest sample is not the randomized contrast. Lee (2009) trimming bounds were reported for each training group. These assume that assignment to a training group can lower but not raise any person's probability of completing the posttest. Under that assumption, the control group's posttest-sample distribution is a mixture of participants who would have completed under either assignment and participants who completed only because they were in the control condition. Trimming the control group's distribution from below and from above by the known excess proportion p (the control retention rate minus the training retention rate, divided by the control retention rate) bounds the distribution of the always-completing stratum. The bounds are stated and proved for the mean (Lee, 2009). The always-completing stratum is a principal stratum (Zhang & Rubin, 2003), as used in the trimming problem (Lee, 2009). The extension to quantiles follows from the sharp bounds on the quantiles of a mixture component with known mixing proportion (Horowitz & Manski, 1995); the derivation is in the supplement under Lee Bounds for Detachment. The gradient bound is the cross-combination of the quantile bounds and is valid but not necessarily sharp. Intervals follow Imbens and Manski (2004). The monotonicity

assumption was stated as an assumption, and bounds were also reported at 1.5p in case p is understated, and within baseline terciles as a tightening (Semenova, 2025). Multiple imputation by chained equations (predictive mean matching, $m = 50$, 20 iterations; van Buuren & Groothuis-Oudshoorn, 2011) imputed each posttest outcome from the baseline scales, the occasion-1 item parcels, and the two condition indicators. Estimates were combined by Rubin's (1987) rules, with bootstrap draws pooled across imputations for the quantile intervals. A delta-adjusted sensitivity shifted imputed values in the training groups by −0.25 and −0.50 scale points to represent dropouts who would have scored lower.

Dispersion and Distribution Regression

The log ratio of the posttest standard deviation in each training group to that in the control group was reported with a bootstrap interval and a Brown–Forsythe test. Distribution regression (Chernozhukov et al., 2013) fitted, at each support value y from 2 to 6, a logistic regression of the indicator that posttest detachment is at or below y on condition and centered baseline, giving condition-specific distribution functions at the mean baseline. It was reported as a figure and carries no decision.

Follow-Up Repeat

The quantile treatment effects were repeated on the follow-up parcel mean for the 204 participants with both parcels. This is a descriptive repeat on a different score and sample, not a test of persistence.

Software and Generative Models

All estimates were computed in R (R Core Team, 2025). HC2 standard errors were computed with the sandwich package (Zeileis et al., 2020), quantile regressions with quantreg (Koenker, 2025), and multiple imputations with mice (van Buuren & Groothuis-Oudshoorn,

2011); the changes-in-changes estimates were cross-checked against an independent implementation (Callaway, 2026), package versions were fixed with renv (Ushey & Wickham, 2026), results were stored with jsonlite (Ooms, 2014), and figures were drawn with ggplot2 (Wickham, 2016). The Lee bounds, the recentered-influence-function quantiles, and the discrete changes-in-changes bounds were coded for this study and unit-tested. The manuscript was converted to a word-processor file with pandoc (MacFarlane, 2026) and Python (Python Software Foundation, 2025). Large language models, used under my direction through Cursor (Anysphere, 2026) and the Claude application, drafted and revised text, wrote and reviewed analysis and figure code, and checked reported statistics: Claude Fable 5.1 (Anthropic, 2026a), Claude Haiku 5.5 (Anthropic, 2026b), Claude Opus 5.5 (Anthropic, 2026c), Claude Sonnet 5.5 (Anthropic, 2026d), Grok 4.7 (Anysphere & SpaceXAI, 2026), GPT-5.6 Sol (OpenAI, 2026), Kimi K3 (Moonshot AI, 2026), Gemini 3.1 Pro (Google, 2026), and Composer 2.5 (Anysphere, 2026). No model received respondent-level data in conversation. I set the questions, the plan, every decision rule, and every decision, and I am responsible for the content. The supplemental material gives the full automation record, including the role of each model and tool.

Data, Materials, and Code

The deposited data and study materials are available at the Open Science Framework (https://osf.io/32rhy/; Peuckmann et al., 2026); respondent-level files are not redistributed here. The supplemental material is available at https://osf.io/r68v7/. The analysis code is not part of the supplemental material.

Precision

Because the trial was not designed for this question, the precision of each quantity was studied by simulation before the posttest data were analyzed. The simulation used the observed

cell sizes and attrition rates and the published baseline moments, across a grid that included the null (63 cells, 200 replicates each). For a baseline × condition interaction of ±0.5 scale points of gain per scale point of baseline (about ±0.69 per baseline standard deviation), the Lin interaction reached power of .72 to .80 at a latent pre–post correlation of .45 and .79 to .93 at .70, with Type I error at .05. At ±0.3 per scale point, power was .33 to .57. The quantile gradient was far less precise. A baseline slope of −0.5 produces a marginal gradient of only −0.21 (at ρ = .45) to −0.45 (at ρ = .70), because a baseline-conditional effect is attenuated in the marginal posttest distribution. The gradient's 95% interval averaged 1.6 to 1.8 scale points in width, and power reached only .65 for a marginal gradient of a full scale point. The log standard-deviation ratio and the Brown–Forsythe test had power above .80 only for a marginal gradient of a scale point. For H4, the missingness interaction reached power of .48 for a dependence of −0.5 logits per baseline standard deviation confined to the training groups and .87 for −0.8. These figures were the reason the Lin interaction was fixed as the primary test before the freeze, and they are the scale against which the intervals below should be read. A moderate baseline-conditional effect is within what this design can detect, and a modest marginal gradient is within what it can only bound.

Results

Sample, Retention, and Baseline

Table 1 gives the sample sizes, retention, and the depositors' published descriptive statistics by condition. Of the 393 randomized participants, 247 (62.8%) provided posttest recovery scores; loss was 30.3% in control, 39.7% in CBT, and 41.5% in MBT. Within each group, baseline detachment did not detectably differ in distribution between participants who did and did not provide a posttest score (Kolmogorov–Smirnov $D = .16$, $p = .466$ in control; $D = .08$,

p = .983 in CBT; D = .11, p = .835 in MBT). Baseline detachment was low in all three groups (pooled M = 2.72, SD = 1.37 on the 1–7 scale; Peuckmann et al., 2026). By chance, it was higher in CBT (M = 3.00) than in control (M = 2.53) or MBT (M = 2.63; Peuckmann et al., 2026, Table S1), a difference of about a third of a standard deviation that the Lin adjustment addresses.

Family A: Baseline × Condition Interactions on Detachment (H1–H3)

Table 2 reports the Lin model on the posttest sample (n = 247; R^2 = .32). At the mean baseline in the posttest sample (2.76), posttest detachment was 0.89 scale points higher in CBT than in control (95% CI [0.47, 1.30]) and 0.98 higher in MBT (95% CI [0.57, 1.39]). In control, posttest detachment rose 0.60 scale points per scale point of baseline (95% CI [0.38, 0.82]).

The baseline × CBT interaction was −0.21 (SE = 0.15, 95% CI [−0.50, 0.08], 99% CI [−0.59, 0.17], p = .163; Holm-adjusted p = .488). Posttest detachment therefore rose more slowly with baseline in CBT (0.39 per scale point) than in control (0.60), the direction compensation predicts. The interval excludes an accumulation slope larger than 0.08 and a compensation slope more negative than −0.50 scale points per scale point of baseline; because it also covers zero, the result is indeterminate under the decision rule. The baseline × MBT interaction was 0.04 (SE = 0.15, 95% CI [−0.24, 0.33], 99% CI [−0.33, 0.42], p = .764; Holm-adjusted p = 1.000), indeterminate and centered near zero. The slope in MBT (0.64) was almost identical to that in control. H1 and H2 are therefore indeterminate, and neither compensation nor accumulation is supported for either training.

The H3 contrast, MBT minus CBT, was 0.25 (SE = 0.13, 95% CI [−0.01, 0.51], 99% CI [−0.09, 0.59], p = .061). Its sign is the one the account-specific predictions imply (a more positive interaction in MBT than in CBT), but the interval includes zero and the contrast is indeterminate under the decision rule.

Family B: Where on the Distribution the Change Registered

Quantile Treatment Effects

Figure 1 gives the fitted distribution functions by condition from the distribution regression reported below; Table 3 and Figure 2 give the quantile treatment effects on posttest detachment in the posttest sample. The control group's quartiles were 2.00, 3.25, and 4.56. Relative to control, the CBT quantiles were higher by 1.50 at $\tau = .25$ (95% CI [0.75, 2.00]), 1.25 at $\tau = .50$ (95% CI [0.50, 2.00]), and 1.06 at $\tau = .75$ (95% CI [0.00, 2.00]). The MBT quantiles were higher by 1.44 (95% CI [0.50, 2.00]), 1.38 (95% CI [0.75, 2.00]), and 1.19 (95% CI [0.31, 2.00]). In both training groups the point estimates decline from the .25 to the .75 quantile. The gradients were −0.44 for CBT (95% CI [−1.50, 0.56]; BCa [−1.50, 0.50]; bootstrap $p = .542$) and −0.25 for MBT (95% CI [−1.13, 0.88]; BCa [−1.38, 0.50]; $p = .907$). Both are indeterminate, and both intervals are about two scale points wide. The difference between gradients (MBT minus CBT) was 0.19 (95% CI [−0.50, 1.13]). At the descriptive tails, the CBT effect at $\tau = .10$ was 1.25 (95% CI [0.02, 1.75]) and the MBT effect 0.38 (95% CI [−0.25, 1.58]). At $\tau = .90$ both were about 0.75 to 0.80, where the training groups' .90 quantiles (6.50 and 6.55) sit close to the scale maximum of 7. Nine (CBT), 10 (MBT), and 11 (control) distinct support values fell between the .25 and .75 quantiles. The baseline-adjusted recentered-influence-function gradient was −0.95 for CBT (percentile 95% CI [−1.86, 0.09]; BCa [−1.97, −0.04]) and −0.03 for MBT (95% CI [−1.27, 0.96]). The CBT percentile and BCa intervals disagree about zero, and the prespecified decision interval is the percentile interval. Kernel-smoothed quantiles gave the same ordering.

Changes-in-Changes

Changes-in-changes uses the control group's own pre-to-post movement as the counterfactual; the control quantiles rose 0.50 at $\tau = .25$, 1.00 at $\tau = .50$, and 1.06 at $\tau = .75$. The

discrete-outcome bounds on the CBT effect in the posttest sample were [1.00, 1.25] at τ = .25, [0.50, 0.50] at τ = .50, and [0.25, 0.50] at τ = .75, giving a gradient bound of [−1.00, −0.50] (jittered point estimate −0.62). For MBT the bounds were [0.75, 1.00], [0.75, 1.25], and [0.75, 1.25], with a gradient bound of [−0.25, 0.50] (jittered 0.01). The repeated-cross-section form gave gradient bounds of [−1.25, −0.50] for CBT and [−0.75, −0.25] for MBT. The estimates agreed with an independent implementation of the estimator to within 0.19 scale points at every quantile. Changes-in-changes thus points the same way as the quantile effects for CBT, toward larger shifts at lower quantiles, and is uninformative about sign for MBT in the panel form. The plan specified no bootstrap interval for the bounds, and they are reported as identification bounds, not as tests.

Bounds for Differential Attrition

Table 4 gives the Lee bounds. With trimming fractions of p = .135 (CBT) and .161 (MBT), the mean effect among participants who would have completed under either assignment was bounded in [0.76, 1.48] for CBT and [0.65, 1.48] for MBT, with Imbens–Manski 95% intervals of [0.27, 1.99] and [0.12, 2.02]. The gradient bounds were [−0.88, 0.88] for CBT and [−0.69, 1.31] for MBT; at 1.5p they were [−0.88, 1.38] and [−1.06, 1.56]; within baseline terciles, [−0.88, 0.50] and [−0.69, 0.88]. Every gradient bound straddles zero. The bounds indicate that differential attrition of the observed size could move the gradient by up to about a scale point in either direction; they do not alter the Family A readings, which remain indeterminate.

Multiple Imputation

Under imputation from baseline given condition, the Lin condition effects at the mean baseline were 0.58 for CBT (95% CI [−0.06, 1.21]; fraction of missing information λ = .53) and

0.64 for MBT (95% CI [−0.10, 1.39]; λ = .65), smaller than the posttest-sample estimates and with intervals that reach zero. Shifting imputed training-group values by −0.25 and −0.50 reduced them further (CBT 0.48 and 0.38; MBT 0.54 and 0.44). The imputed quantile effects for CBT were 0.69 (95% CI [−0.25, 2.00]), 1.27 (95% CI [0.00, 2.25]), and 0.84 (95% CI [−0.13, 2.00]) at the three quartiles. The imputed gradients were 0.16 for CBT (95% CI [−1.75, 1.63]) and 0.33 for MBT (95% CI [−1.75, 1.88]), with between-imputation standard deviations of about 1.0, so that the imputation intervals for the gradient are uninformative. Chain means of imputed detachment showed no trend across 20 iterations (range 3.51–4.85; supplement Table S19).

Dispersion

The posttest standard deviation of detachment was 1.56 in control, 1.48 in CBT, and 1.65 in MBT (Peuckmann et al., 2026, Table S1). The log ratio to control was −0.06 for CBT (95% CI [−0.25, 0.12]; Brown–Forsythe p = .431) and 0.05 for MBT (95% CI [−0.12, 0.22]; p = .673). Neither training group's posttest distribution was detectably narrower or wider than the control group's.

Distribution Regression

Figure 1 shows the fitted distribution functions by condition at the mean baseline. The curves for the two training groups lie below the control curve at every support value, by a roughly constant vertical distance in the middle of the scale, which is consistent with a location shift rather than a compression or spread. The implied quartiles were 2.75, 3.50, and 4.25 (control), 3.50, 4.50, and 5.25 (CBT), and 3.75, 4.75, and 5.75 (MBT). The implied gradients were 0.25 (CBT) and 0.50 (MBT), of opposite sign to the type-7 gradients.

Family A: Was Attrition Patterned by Baseline Detachment (H4)?

Table 5 reports the missingness model on all 393 participants. In control, the log-odds of missing the posttest fell by 0.14 per scale point of baseline detachment (95% CI [−0.45, 0.16]; −0.20 per baseline *SD*); that is, lower-baseline participants were slightly more likely to be lost, but the interval covers zero. The slopes were −0.04 in CBT (95% CI [−0.30, 0.22]) and −0.07 in MBT (95% CI [−0.33, 0.18]). The baseline × condition interactions were 0.11 for CBT (95% CI [−0.30, 0.51]) and 0.07 for MBT (95% CI [−0.33, 0.47]); the 2-*df* Wald test gave $\chi^2(2) = 0.27$, $p = .875$ (Holm-adjusted $p = 1.000$). H4 is indeterminate. Attrition was higher in the training groups (log-odds 0.46 and 0.52 higher than control at the mean baseline, with intervals that touch zero), but the baseline dependence of attrition did not detectably differ by condition, and the Kolmogorov–Smirnov comparisons above agree. Neither the compensation prediction (that CBT retains low-baseline participants at least as well as the control condition) nor the accumulation prediction (that low-baseline participants are lost at a higher rate from the training groups) is supported.

Secondary Outcomes

The Lin model on each secondary outcome (supplement Table S3) showed condition effects at the mean baseline of 0.45 (CBT; 95% CI [0.07, 0.84]) and 0.33 (MBT; 95% CI [−0.07, 0.73]) on relaxation, 0.37 (95% CI [0.01, 0.74]) and 0.26 (95% CI [−0.10, 0.61]) on control, and 0.40 (95% CI [0.01, 0.79]) and 0.54 (95% CI [0.14, 0.94]) on mastery. On the standardized scores, the effects were 0.32 (95% CI [0.06, 0.58]) and 0.21 (95% CI [−0.04, 0.45]) for sleep quality and −0.38 (95% CI [−0.64, −0.12]) and −0.26 (95% CI [−0.54, 0.02]) for perceived stress. Of the 10 baseline × condition interactions on secondary outcomes, nine had intervals covering zero. The exception was baseline detachment × CBT on perceived stress, −0.28 (95%

CI [−0.54, −0.02], p = .033; 99% CI [−0.62, 0.06]), which is one unadjusted test among 10 and is not interpreted. Quantile treatment effects and gradients on relaxation, control, and mastery (Figure S1) were small, with gradients between −0.44 and 0.50 and every interval covering zero. The log standard-deviation ratios on these outcomes likewise had intervals covering zero (Brown–Forsythe p values from .29 to .97).

Follow-Up Repeat (Descriptive)

On the follow-up parcel mean (n = 204), the CBT quantile effects were 1.06 (95% CI [0.00, 1.75]), 0.63 (95% CI [0.13, 1.50]), and 0.75 (95% CI [−0.25, 1.38]) at the three quartiles (gradient −0.31, 95% CI [−1.25, 0.56]). The MBT effects were 0.06 (95% CI [−0.75, 1.25]), 0.63 (95% CI [−0.13, 1.50]), and 0.25 (95% CI [−0.50, 1.19]) at the three quartiles (gradient 0.19, 95% CI [−1.25, 1.13]).

Summary of Distributional and Bounds Evidence

The Lin benchmark says that both trainings raised detachment at the mean baseline by about 0.9 scale points and differed from each other by 0.09, a difference for which the plan included no test. The distributional analyses add four statements, each with a bound attached. First, neither training detectably compressed or spread the posttest distribution. Second, the baseline dependence of each training's effect was indeterminate, within intervals of about 0.6 scale points per scale point of baseline. Third, attrition was higher in the training groups but not detectably patterned by baseline in any group. Fourth, the Lee bounds put the mean effect among participants who would have completed under either assignment between about 0.65 and 1.5 scale points, while imputation under a missing-at-random assumption gave mean effects of about 0.6 with intervals that reached zero. Under the prespecified contribution criterion, none of these results changes the conclusion the mean contrast supports or the published mean-equivalence

finding (Reis et al., 2024). What they add is the range of compensation or accumulation that remains compatible with the data.

Discussion

The present study asked whether two recovery trainings that raise mean detachment did so by helping employees who detached least (compensation) or employees who already detached well (accumulation). Compensation and accumulation were set against each other, condition by condition, in a prespecified test on a deposited three-group randomized trial, and under the decision rule the data support neither account for either training. In CBT the interaction leaned toward compensation (H1); in MBT it sat near zero (H2); both intervals covered zero, and the condition contrast (H3) was positive with an interval that reached zero. Quantile effects were larger at the .25 than at the .75 quantile in both training groups, but their intervals spanned up to two scale points. The gradient bounds for differential attrition straddled zero, and attrition itself was not detectably patterned by baseline detachment in any group (H4). Both trainings raised detachment at the mean baseline by about 0.9 points on a 7-point scale, and did so to a similar extent, as the depositing team reported (Reis et al., 2024).

What the Intervals Rule Out

An interval that covers zero still bounds the effect it estimates. The CBT interval excludes an accumulation slope larger than 0.08 scale points of additional gain per scale point of baseline detachment. In this trial, then, CBT did not detectably favor employees who were already detaching well. It also excludes a compensation slope more negative than −0.50 per scale point, so a compensation effect of the size the simulation showed the design could detect with power of .72 to .80 is unlikely. For MBT, the interval excludes slopes more negative than −0.24 and more positive than 0.33. Against the accumulation account’s most confident prediction,

spread under MBT, the MBT interaction of 0.04 is near zero, and its interval excludes a spread steeper than 0.33 per scale point. For the compensation account's most confident prediction, compression under CBT, the CBT interaction of −0.21 is in the predicted direction and about 40% of the slope of 0.5 that the design was powered to detect. On balance, a compensation effect of modest size under CBT cannot be excluded; an accumulation effect of that size is excluded under CBT and not under MBT.

The condition contrast of 0.25 (95% CI [−0.01, 0.51]) should be read with care in both directions. It is the difference the account-specific predictions imply, with the MBT interaction more positive than the CBT interaction, and its interval barely reaches zero. It is also a secondary reading, its *p* value of .061 was not adjusted, and the two interactions it contrasts were each indeterminate. Read as weak evidence that the two trainings differ in whom they help, it calls for replication in a trial designed to test the contrast.

Comparison With Earlier Findings

These results are consistent with what the depositing team and others have reported. The depositing team concluded that the two trainings raised detachment to a similar extent (Reis et al., 2024). The Lin benchmark here agrees, and the distributional analysis finds no case in which equivalent means concealed opposite distributional effects, the scenario warned of in program evaluation (Bitler et al., 2006). A mean gain of about 0.9 scale points, somewhat more than half a standard deviation of the posttest score, is larger than the meta-analytic average for detachment interventions ($d = 0.36$; Karabinski et al., 2021), consistent with that meta-analysis's finding that longer, higher-dose programs produce larger effects. Finding no detectable baseline moderation matches the earlier test of a baseline interaction in a mindfulness recovery intervention, which found none (Hülsheger et al., 2015). It does not contradict the finding that segmentation

preference moderated a mindfulness training's effect on detachment (Althammer et al., 2021). Segmentation preference is a disposition about the work–home boundary, not the baseline level of the outcome, and the present data contain no such moderator.

Across studies, detachment interventions had larger effects among participants with initial impairments (Karabinski et al., 2021), a between-study moderation result that a within-trial analysis could corroborate. On the decision rule, this one did not, although the point estimates lean that way in CBT.

Theoretical Implications

For the aptitude-treatment and cumulative-advantage accounts, the results set bounds on either mechanism rather than deciding between them. Neither account predicted a flat baseline × condition relation; both predicted a sign. Where the accumulation account was most confident, MBT, the data returned an interaction of 0.04 with an interval that runs from −0.24 to 0.33 per scale point. If the cumulative-advantage mechanism operates in self-guided mindfulness training, it did not operate detectably on detachment in this sample, which was composed of volunteers who, by enrolling, had already shown some capacity to act on their own recovery. Compensation fared no better by the decision rule, but the CBT estimate was in the predicted direction, and the design's power at that effect size was limited. A prediction that a skill training helps those who lack the skill most is a claim about a specific skill deficit, and baseline detachment is a crude proxy for lacking the skill of boundary management. The plan said in advance that baseline detachment is not a measure of capacity, and the data add that it is not a strong moderator of either training's effect.

One further point concerns which quantity a study estimates. The two accounts predict baseline-conditional heterogeneity. The marginal quantile treatment effect, which is what

distributional analyses of trials usually report, is an indirect test of that prediction. The simulation showed that a baseline slope of −0.5 produces a marginal gradient of only −0.21 to −0.45 because the baseline-conditional effect is attenuated when it is read off the marginal posttest distribution. Studies that ask for whom a training works should prespecify the baseline-conditional interaction as the primary quantity and treat the marginal quantile effect as a description of where the change registered, as this analysis does.

Practical Implications

The results give no reason to target either training at employees who detach less, or to withhold either from employees who detach more. Both trainings were associated with higher posttest detachment across the distribution in the posttest sample, and neither detectably favored one end. On these results, organizations offering self-guided recovery trainings may offer them broadly. More useful for practice is the finding that attrition was higher in the training groups but not detectably patterned by baseline detachment: the employees most in need of recovery were not disproportionately the ones who dropped out, at least as measured by where they started.

Limitations

Several limits bear directly on the conclusions. The first concerns precision. The trial was sized for a mean comparison, and the distributional estimates have intervals up to two scale points wide, somewhat wider than the 1.6 to 1.8 scale points the simulation anticipated. The paper’s bounds are what the design permits, and a trial designed to test compensation against accumulation would need either a much larger sample or a design that stratifies randomization on baseline. Attrition is the second. Thirty-seven percent of randomized participants did not provide a posttest score, and the rate was higher in the training groups. The contrast within the posttest sample is therefore not the randomized contrast; the Lee bounds and the imputation analysis say

how far that matters. For the mean effect, imputation under a missing-at-random assumption given baseline returned smaller effects with intervals that reached zero, so the effect in the posttest sample of this volunteer trial may overstate what a broad offer would achieve. The third is measurement. Detachment is a four-item self-report on a quarter-point grid, and the grid is coarse where the gradient is read (9 to 11 support values between the quartiles). The distribution-regression gradients, which point in the opposite direction from the type-7 gradients, and the panel changes-in-changes bounds for MBT, which include zero, show how much the gradient depends on the quantile definition on such a grid. The deposited file also contains no measure of self-regulatory capacity, so the trial does not measure the mechanism either account posits. Sample composition is the fourth. Participants were German employees with low baseline detachment who volunteered for recovery training; three quarters were women, health care and education was the largest occupational group, and the trial did not report racial or ethnic composition. The compensation account is therefore tested in a population where nearly everyone has room to gain, and the accumulation account in one where nearly everyone had shown enough initiative to enroll. Whether either training acts differently among employees who do not seek training, among men, or outside a Western European labor market cannot be determined from these data. The fifth is my prior exposure. The plan was frozen after I had seen pooled column aggregates of the posttest scores and the posttest counts by condition. The decision statistics were never computed before the single run, and the run procedure is documented in the supplement; a reader who weighs an analysis plan by the standard of zero prior contact should discount accordingly. The sixth is the baseline imbalance in detachment (CBT 3.00 versus control 2.53). Covariate adjustment is the right remedy in a randomized design, but the imbalance means that the unadjusted quantile effects for CBT include a

component of baseline difference. The adjusted recentered-influence-function gradient for CBT was accordingly more negative than the unadjusted one.

Three considerations also temper any reading of the CBT results as weak support for compensation. One is the prespecified decision rule itself, which says indeterminate. Another is the sample: volunteers with low baseline detachment overall ($M = 2.72$), so the compensatory prediction is tested in a population where almost everyone has room to gain. A third is a ceiling. The training groups' .90 quantiles sit within half a point of the scale maximum (6.50 and 6.55 on the 1–7 scale), so part of the decline across quantiles is a ceiling pattern.

Future Directions

Three studies would resolve what this one leaves open. A recovery-training trial with randomization stratified on baseline detachment and a posttest sample roughly five to six times this size (the square of 0.5/0.21) would test the baseline × condition interaction at the observed effect size with adequate power. That multiple follows from inverse-square scaling of the standard error from the 0.5 per scale point this design could detect to the 0.21 observed. A trial that measures self-regulatory capacity at baseline, rather than inferring it from the baseline outcome, would test the mechanism each account posits rather than its proxy. A trial that uses a brief outcome measure to follow participants who drop out would replace the bounds reported here with an estimate.

Conclusion

Two 6-week recovery trainings raised psychological detachment by about 0.9 points on a 7-point scale in the posttest sample, and did so to a similar extent. A prespecified test of whether either training helped low-baseline or high-baseline employees more returned an indeterminate result for both conditions and for their contrast. The intervals exclude a compensation effect

larger than the design was built to detect in either condition and an accumulation effect of practical size in CBT. Dropout before the posttest was higher in the training groups but was not detectably patterned by baseline detachment. The distributional analysis confirms the mean-level conclusion already published for this trial, and the intervals state the precision a later trial would need to detect what this one could only bound. Until such a trial is run, organizations that offer recovery training would do well to offer it across the range of baseline detachment rather than reserve it for employees who detach least.

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Tables

Table 1

Sample, Retention, and Descriptive Statistics for Psychological Detachment by Condition

Measure	Control	CBT	MBT	Total
Randomized *n*	132	131	130	393
Posttest score present, *n* (%)	92 (69.7)	79 (60.3)	76 (58.5)	247 (62.8)
Follow-up parcels present, *n*	68	67	69	204
Baseline detachment, *M* (*SD*)	2.53 (1.32)	3.00 (1.34)	2.63 (1.42)	2.72 (1.37)
Posttest detachment, *M* (*SD*)	3.46 (1.56)	4.55 (1.48)	4.49 (1.65)	4.13 (1.64)
Posttest detachment, quartiles	2.00 / 3.25 / 4.56	3.50 / 4.50 / 5.63	3.44 / 4.63 / 5.75	
McDonald's ω, detachment, pretest / posttest				.87 / .92
Baseline detachment, with vs. without posttest score, KS *D* (*p*)	.16 (.466)	.08 (.983)	.11 (.835)	

Note. CBT = cognitive-behavioral recovery training; MBT = mindfulness-based training; KS = two-sample Kolmogorov–Smirnov test. Means, standard deviations, and ω are from Peuckmann et al. (2026, Tables 1 and S1); ω is reported there for the pooled sample only. Posttest statistics describe the posttest sample, the participants with a posttest score. Detachment is a four-item mean on a 1–7 scale. Quartiles are type-7 sample quantiles at .25, .50, and .75.

Table 2

Lin-Adjusted Regression of Posttest Detachment on Condition and Baseline Detachment

Term	*b*	*SE*	95% CI	*p*	Holm *p*
Intercept (control at mean baseline)	3.56	0.14	[3.28, 3.84]	< .001	
CBT	0.89	0.21	[0.47, 1.30]	< .001	
MBT	0.98	0.21	[0.57, 1.39]	< .001	
Baseline	0.60	0.11	[0.38, 0.82]	< .001	
Baseline × CBT (H1)	−0.21	0.15	[−0.50, 0.08]	.163	.488
Baseline × MBT (H2)	0.04	0.15	[−0.24, 0.33]	.764	1.000
(Baseline × MBT) − (Baseline × CBT) (H3)	0.25	0.13	[−0.01, 0.51]	.061	

Note. n = 247 (posttest sample). CBT = cognitive-behavioral recovery training; MBT = mindfulness-based training. Estimates are from ordinary least squares with heteroskedasticity-consistent (HC2) standard errors (Lin, 2013). Baseline detachment is centered at the posttest-sample mean (2.76). R^2 = .32. The CBT and MBT coefficients are the mean benchmarks. Holm *p* values are adjusted across the three Family A tests (the two interactions here and the 2-*df* Wald test in Table 5). Under the decision rule fixed in the plan (compensation if the 95% CI lies below zero, accumulation if it lies above zero, indeterminate otherwise), H1, H2, and H3 are indeterminate. The 99% CIs, also fixed in the plan, are [3.20, 3.92] for the intercept, [0.34, 1.44] for CBT, [0.44, 1.52] for MBT, [0.31, 0.89] for baseline, [−0.59, 0.17] for Baseline × CBT, [−0.33, 0.42] for Baseline × MBT, and [−0.09, 0.59] for the contrast.

Table 3

Quantile Treatment Effects on Posttest Detachment, Each Training Group Versus Control

Quantile or summary	CBT estimate	CBT 95% CI	MBT estimate	MBT 95% CI
.10 (descriptive)	1.25	[0.02, 1.75]	0.38	[−0.25, 1.58]
.25	1.50	[0.75, 2.00]	1.44	[0.50, 2.00]
.50	1.25	[0.50, 2.00]	1.38	[0.75, 2.00]
.75	1.06	[0.00, 2.00]	1.19	[0.31, 2.00]
.90 (descriptive)	0.80	[0.08, 1.50]	0.75	[0.10, 1.25]
Gradient, QTE(.75) − QTE(.25)	−0.44	[−1.50, 0.56]	−0.25	[−1.13, 0.88]
Gradient, BCa interval		[−1.50, 0.50]		[−1.38, 0.50]
Gradient, baseline-adjusted RIF	−0.95	[−1.86, 0.09]	−0.03	[−1.27, 0.96]

Note. Posttest sample: control $n = 92$, CBT $n = 79$, MBT $n = 76$. CBT = cognitive-behavioral recovery training; MBT = mindfulness-based training; QTE = quantile treatment effect (difference of type-7 sample quantiles, training group minus control group); RIF = recentered influence function regression adjusted for centered baseline detachment and its products with the condition indicators; BCa = bias-corrected and accelerated. Intervals are percentile bootstrap intervals (participants resampled within condition, $B = 2{,}000$, seed 20261022) unless labeled BCa. A negative gradient means a larger effect at the .25 than at the .75 quantile. The difference of gradients, MBT − CBT, was 0.19, 95% CI [−0.50, 1.13]. The discrete changes-in-changes bounds on the gradient were [−1.00, −0.50] for CBT and [−0.25, 0.50] for MBT (supplement Table S10). The number of distinct support values between the .25 and .75 quantiles is 9 (CBT), 10 (MBT), and 11 (control). All gradient readings are indeterminate under the decision rule.

Table 4

Lee Bounds and Multiple-Imputation Estimates for Posttest Detachment Under Differential Attrition

Estimate	CBT	MBT
Lee trimming fraction *p*	.135	.161
Mean effect, Lee bounds	[0.76, 1.48]	[0.65, 1.48]
Mean effect, Imbens–Manski 95% CI	[0.27, 1.99]	[0.12, 2.02]
QTE(.25), Lee bounds	[0.75, 1.50]	[0.69, 1.44]
QTE(.75), Lee bounds	[0.63, 1.63]	[0.75, 2.00]
Gradient, Lee bounds	[−0.88, 0.88]	[−0.69, 1.31]
Gradient, Imbens–Manski 95% CI	[−1.79, 2.04]	[−1.59, 2.46]
Gradient, Lee bounds at 1.5*p*	[−0.88, 1.38]	[−1.06, 1.56]
Gradient, Lee bounds within baseline terciles	[−0.88, 0.50]	[−0.69, 0.88]
Mean effect at mean baseline, MI	0.58	0.64
Mean effect, MI 95% CI	[−0.06, 1.21]	[−0.10, 1.39]
Mean effect, MI, $\delta = -0.25$	0.48	0.54
Mean effect, MI, $\delta = -0.50$	0.38	0.44
QTE(.25), MI	0.69	0.64
QTE(.50), MI	1.27	1.24
QTE(.75), MI	0.84	0.97
Gradient, MI	0.16	0.33
Gradient, MI 95% CI	[−1.75, 1.63]	[−1.75, 1.88]
Gradient, MI, $\delta = -0.25$	0.21	0.34
Gradient, MI, $\delta = -0.50$	0.25	0.33

Note. Each column is one training group versus control. CBT = cognitive-behavioral recovery training; MBT = mindfulness-based training; QTE = quantile treatment effect; MI = multiple imputation by chained equations (predictive mean matching, $m = 50$, 20 iterations; predictors: baseline scales, occasion-1 parcels, condition indicators), combined by Rubin's rules; δ = shift applied to imputed training-group values. Lee bounds trim the control group's posttest-sample distribution by *p*, separately from below and from above, under the assumption that training assignment can only lower the probability of completing the posttest. The gradient bound is the cross-combination of the quantile bounds; it is valid but not necessarily sharp. The fraction of missing information for the MI mean effects is $\lambda = .53$ (CBT) and .65 (MBT).

Table 5

Logistic Regression of Posttest Missingness on Baseline Detachment and Condition

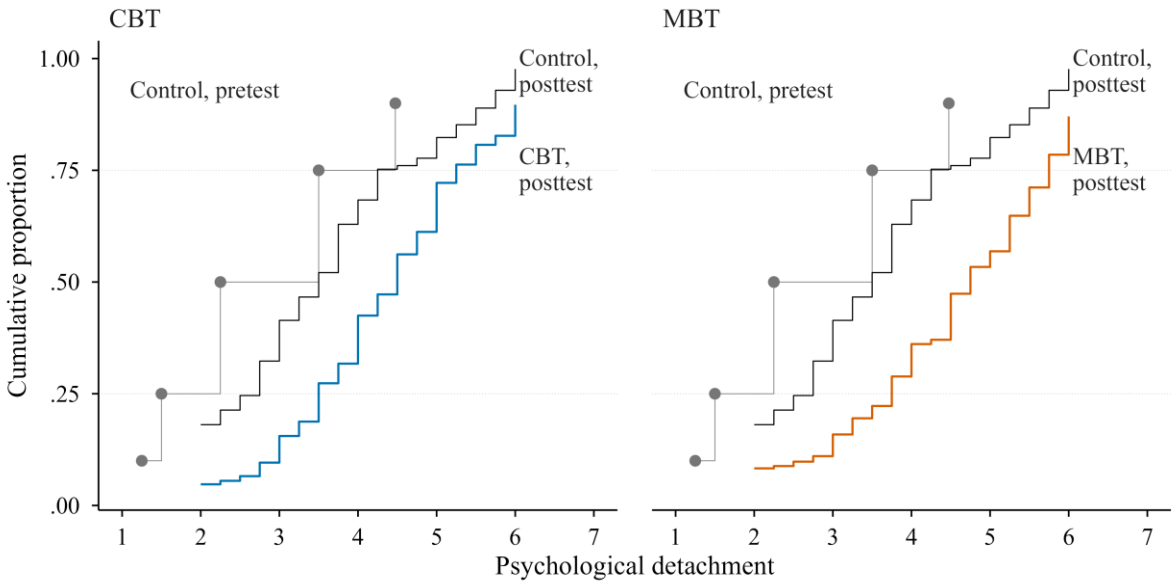
Term	*b*	95% CI	*p*
Intercept (control at mean baseline)	−0.87	[−1.25, −0.48]	< .001
Baseline (slope in control)	−0.14	[−0.45, 0.16]	.354
CBT	0.46	[−0.07, 0.99]	.088
MBT	0.52	[−0.01, 1.04]	.052
Baseline × CBT	0.11	[−0.30, 0.51]	.608
Baseline × MBT	0.07	[−0.33, 0.47]	.733
2-*df* Wald test of the two interactions (H4)	$\chi^2(2) = 0.27$		.875
Implied slope in CBT	−0.04	[−0.30, 0.22]	
Implied slope in MBT	−0.07	[−0.33, 0.18]	

Note. $N = 393$. CBT = cognitive-behavioral recovery training; MBT = mindfulness-based training. Coefficients are log-odds with heteroskedasticity-consistent (HC2) standard errors; positive values indicate higher odds of missing the posttest. Baseline detachment is centered at the full-sample mean ($SD = 1.37$). Per baseline *SD*, the slope in control is −0.20 (95% CI [−0.62, 0.22]) and the interactions are 0.15 (95% CI [−0.41, 0.70]) for CBT and 0.09 (95% CI [−0.45, 0.64]) for MBT (supplement Table S25). The posttest missing rate is 37.2%. The Holm-adjusted *p* for the Wald test is 1.000. H4, read from the two interaction intervals, is indeterminate.

Figures

Figure 1

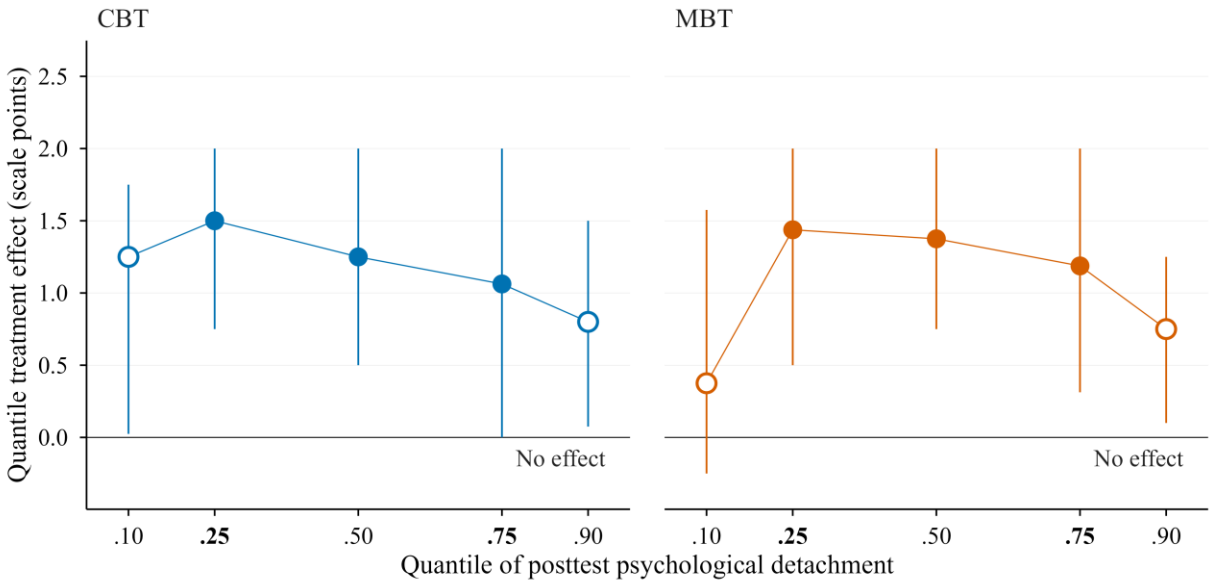
Distribution Functions of Psychological Detachment by Condition



Note. Step functions are fitted posttest distribution functions from distribution regression at the mean baseline score, evaluated at posttest scores from 2.00 to 6.00 in steps of 0.25 (scale range 1 to 7). Each panel compares one training group with the control group. Gray points are the control group's pretest quantiles at .10, .25, .50, .75, and .90 (completers, $n = 92$); assignment to condition was random. A curve to the right means that fewer participants scored at or below a given level, that is, more detachment. Dotted lines mark cumulative proportions of .25 and .75. Interval estimates for the condition differences are shown in Figure 2. CBT = cognitive-behavioral recovery training; MBT = mindfulness-based training.

Figure 2

Quantile Treatment Effects on Posttest Psychological Detachment by Condition



Note. Points are differences between each training group and the control group in posttest detachment quantiles at .10, .25, .50, .75, and .90, in scale points (completers: control $n = 92$, CBT $n = 79$, MBT $n = 76$). Vertical lines are pointwise 95% percentile bootstrap confidence intervals ($B = 2{,}000$); they are not simultaneous across quantiles. Thin lines connect the estimates for readability; no estimate lies between quantiles. The horizontal line marks no effect. Filled points are the decision quantiles (.25, .50, .75); open points are descriptive. Bold labels mark .25 and .75, the quantiles that define the gradient. CBT = cognitive-behavioral recovery training; MBT = mindfulness-based training.

Supplemental Material

The online supplement (Tables S1 to S39 and Figure S1) contains the confirmatory run record; full Lin coefficient tables for the five secondary outcomes; and quantile treatment effects, gradients, recentered-influence-function and kernel sensitivities, and distinct-value diagnostics for the secondary outcomes. It also contains the changes-in-changes bounds, control-group

movement, and package cross-check; the Lee bounds under every trimming variant with the quantile-bound derivation; the multiple-imputation estimates, fractions of missing information, delta sensitivities, and chain diagnostics; dispersion and distribution-regression tables; the full missingness model; the follow-up repeat; the precision simulation by cell; the list of dated amendments; the hash manifest; the prior-exposure record; and the full software and automation record. The frozen study plan and the analysis code are not part of the online supplement.